INCH-POUND

A-A-59654B
8 September 2011
SUPERSEDING
A-A-59654A
22 May 2006

COMMERCIAL ITEM DESCRIPTION

BEARINGS, ROLLER, TAPERED, DOUBLE ROW, NORMAL ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP (TYPE TDO)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers government acquisition requirements for complete (cone with rollers and cup) double row, tapered, roller bearings with normal angle, two single cones, and one double cup (type TDO) for general purpose use. These bearings are not intended for use in special precision applications such as on aircraft, precision ordnance, or submarine equipment.

2. CLASSIFICATION. The roller bearings shall be of one type (TDO) and classified by the size codes listed in table I. The column headings in table I refer to bearing characteristics defined in figure 1.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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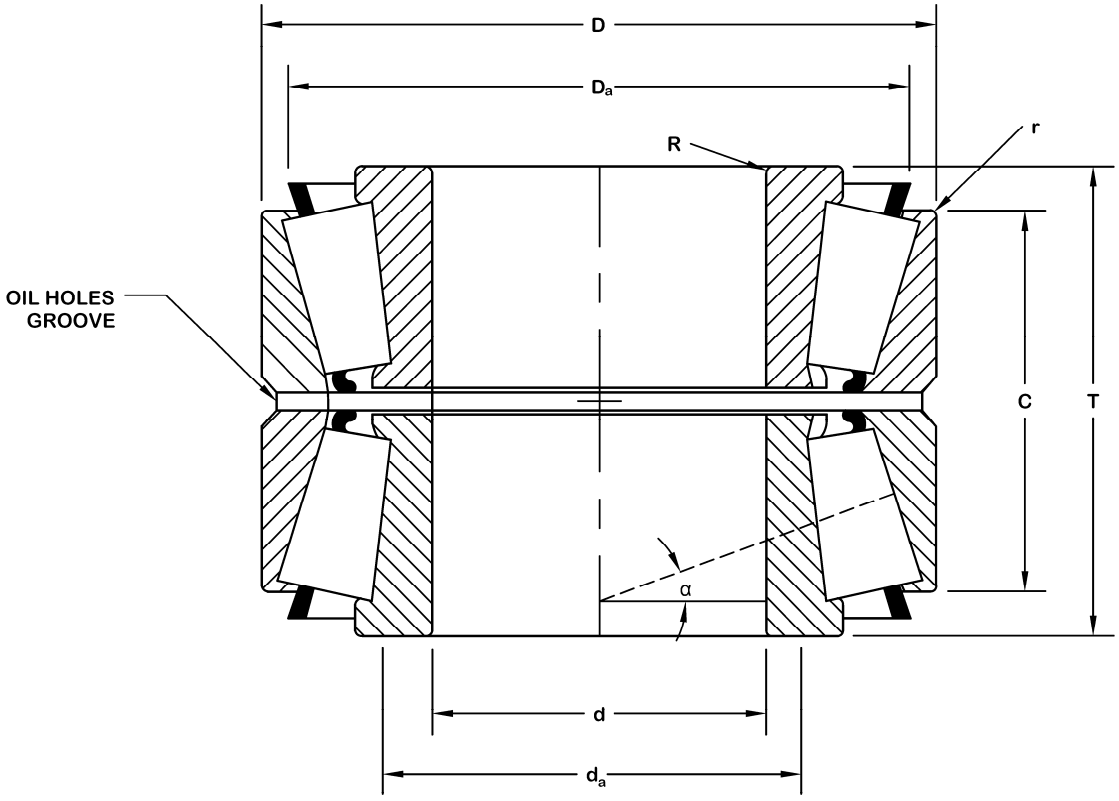


FIGURE 1. Bearing characteristics.

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TABLE I. Size codes and dimensions.

Size code	Part number [1]	d	D	T	B	R [2]	r [2]	d_a	D_a	K factor	Basic dynamic load ratings (lb.)
	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
001	A2037 - A2120D	0.3750	1.2000	0.8370	1.0000	0.05	SPCL	0.59	1.10	1.10	3200
002	A4050 - A4138D	0.5000	1.3775	0.8125	0.9911	0.05	0.03	0.73	1.24	1.24	3700
003	05062 - 05185D	0.6250	1.8504	0.9926	1.2500	0.06	0.03	0.93	1.67	1.64	7500
004	05075 - 05185D	0.7500	1.8504	0.9926	1.2500	0.05	0.03	0.98	1.67	1.64	7500
005	07087 - 08196D	0.8750	1.9687	1.0000	1.3126	0.05	0.03	1.12	1.83	1.45	8100
006	07093 - 07196D	0.9375	1.9687	1.0000	1.3126	0.06	0.025	1.20	1.83	1.45	8100
007	07100 - 07196D	1.0000	1.9687	1.0000	1.3126	0.04	0.025	1.20	1.83	1.45	8100
008	07100S - 07196D	1.0000	1.9686	1.0000	1.3126	0.06	0.025	1.24	1.83	1.45	8100
009	15100 - 15251D	1.0000	2.5000	1.4375	1.8125	0.14	0.03	1.50	2.32	1.67	14000
010	15101 - 15251D	1.0000	2.5000	1.4375	1.8125	0.03	0.03	1.28	2.32	1.67	14000
011	15102 - 15251D	1.0000	2.5000	1.4375	1.8125	0.06	0.03	1.34	2.32	1.67	14000
012	15106 - 15251D	1.0625	2.5000	1.4375	1.8125	0.03	0.03	1.32	2.32	1.67	14000
013	15112 - 15251D	1.1250	2.5000	1.4375	1.8125	0.14	0.03	1.57	2.32	1.67	14000
014	15113 - 15251D	1.1250	2.5000	1.4375	1.8125	0.03	0.03	1.36	2.32	1.67	14000
015	2578 - 2523D	1.1250	2.7500	2.2500	2.6250	0.09	0.03	1.54	2.52	2.14	23100
016	26112 - 26284D	1.1250	2.8338	1.4375	1.6835	0.06	0.03	1.46	2.56	1.62	16300
017	02872 - 02823D	1.1250	3.0000	1.5000	1.8750	0.03	0.03	1.48	2.76	1.29	18200
018	17119 - 17245D	1.1875	2.4409	1.4275	1.5625	0.06	0.03	1.46	2.24	1.53	12000
019	2558 - 2523D	1.1875	2.7500	2.2500	2.6250	0.09	0.03	1.57	2.52	2.14	23100
020	15118 - 15251D	1.1895	2.5000	1.4375	1.8125	0.14	0.015	1.48	2.17	1.67	14000
021	08125 - 08231D	1.2500	2.3125	0.9688	1.2812	0.04	0.03	1.63	2.32	1.23	8900
022	15123 - 15251D	1.2500	2.5000	1.4375	1.7425	SPCL	0.03	1.67	2.32	1.67	14000

[1] Part numbers are for reference only. Part numbers reflect the cup and cone numbers used by industry and the ABMA. Some cones and cups may be used in multiple configurations and are not limited to one bearing assembly.
[2] These maximum fillet radii shall be cleared by the bearing corners.
[3] Minus value indicates load center inside cone backface.

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TABLE I. Size codes and dimensions.

Size code	Part number [1]	d	D	T	B	R [2]	r [2]	d_a	D_a	K factor	Basic dynamic load ratings (lb.)
	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
023	15125 - 15251D	1.2500	2.5000	1.4375	1.8125	0.14	0.03	1.67	2.32	1.67	14000
024	15126 - 15251D	1.2500	2.5000	1.4375	1.8125	0.03	0.03	1.46	2.32	1.67	14000
025	14124 - 14276D	1.2500	2.7170	1.5000	1.8126	0.03	0.03	1.52	2.48	1.53	15100
026	14125A - 14276D	1.2500	2.7170	1.5000	1.8126	0.14	0.03	1.73	2.48	1.53	15100
027	2580 - 2523D	1.2500	2.7500	2.2500	2.6250	0.03	0.03	1.52	2.52	2.14	23100
028	2582 - 2523D	1.2500	2.7500	2.2500	2.6250	0.14	0.03	1.73	2.52	2.14	23100
029	02875 - 02823D	1.2500	3.0000	1.5000	1.8750	0.14	0.03	1.79	2.76	1.29	18200
030	02876 - 02823D	1.2500	3.0000	1.5000	1.8750	0.03	0.03	1.57	2.76	1.29	18200
031	3476 - 3423D	1.2500	3.2500	2.1875	2.6251	0.05	0.03	1.69	2.95	1.60	28900
032	26126 - 26284D	1.2600	2.8338	1.4375	1.6835	0.06	0.03	1.56	2.56	1.62	16300
033	14130 - 14276D	1.3125	2.7170	1.5000	1.8126	0.14	0.03	1.77	2.48	1.53	15100
034	14131 - 14276D	1.3125	2.7170	1.5000	1.8126	0.03	0.03	1.56	2.48	1.53	15100
035	2581 - 2523D	1.3125	2.7500	2.2500	2.6250	0.03	0.03	1.56	2.52	2.11	23100
036	2585 - 2523D	1.3125	2.7500	2.2500	2.6250	0.14	0.03	1.77	2.52	2.14	23100
037	26131 - 26282D	1.3125	2.8125	1.4375	1.6875	0.14	0.02	1.75	2.56	1.62	16300
038	26131 - 36284D	1.3125	2.8338	1.4375	1.6835	0.14	0.03	1.75	2.56	1.62	16300
039	14137A - 14276D	1.3750	2.7170	1.5000	1.8126	0.06	0.03	1.65	2.48	1.53	15100
040	14138A - 14276D	1.3750	2.7170	1.5000	1.8126	0.14	0.03	1.81	2.48	1.53	15100
041	02877 - 02823D	1.3750	3.0000	1.5000	1.8750	0.14	0.03	1.91	2.76	1.29	18200
042	02878 - 02823D	1.3750	3.0000	1.5000	1.8750	0.03	0.03	1.67	2.76	1.29	18200
043	3478 - 3423D	1.3750	3.2500	2.1875	2.6251	0.14	0.03	1.97	2.95	1.60	28800
044	449 - 432D	1.3750	3.7500	2.0000	2.4376	0.03	0.03	1.73	3.43	2.05	35200

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
045	14139 - 14276D	1.3770	2.7170	1.5000	1.8126	0.05	0.03	1.63	2.48	1.53	15100
046	13682 - 13621D	1.4375	2.7170	1.5000	1.8124	0.14	0.03	1.89	2.56	1.45	15400
047	3479 - 3423D	1.4375	3.2500	2.1875	2.6251	0.03	0.03	1.79	2.95	1.60	28800
048	25570 - 25520D	1.4375	3.2650	1.8750	2.2500	0.14	0.03	2.01	3.03	1.74	24900
049	13685 - 13621D	1.5000	2.7170	1.5000	1.8124	0.14	0.03	1.95	2.56	1.45	15400
050	13687 - 13621D	1.5000	2.7170	1.5000	1.8124	0.08	0.03	1.83	2.56	1.45	15400
051	28150 - 28318D	1.5000	3.1510	1.3750	1.8126	0.06	0.03	1.79	2.87	1.45	17700
052	27880 - 27820D	1.5000	3.1510	1.7700	2.2500	0.03	0.03	1.89	2.95	1.04	21700
053	27881 - 27820D	1.5000	3.1510	1.7700	2.2500	0.14	0.03	2.09	2.95	1.04	21700
054	3490 - 3423D	1.5000	3.2500	2.1875	2.6251	0.14	0.03	2.05	2.95	1.60	28800
055	25572 - 25520D	1.5000	3.2650	1.8750	2.2500	0.03	0.03	1.81	3.03	1.74	24900
056	440 - 432D	1.5000	3.7500	2.0000	2.4376	0.03	0.03	1.83	3.43	2.05	35200
057	33880 - 33821D	1.5000	3.7500	2.0624	2.5000	0.14	0.03	2.13	3.54	1.77	35800
058	542 - 533D	1.5000	4.3750	2.5000	3.1250	0.14	0.06	2.17	3.94	1.97	47400
059	28158 - 28318D	1.5748	3.1510	1.3750	1.8126	0.06	0.03	1.87	2.87	1.45	17700
060	24780 - 24720D	1.6250	3.0000	1.5625	1.9375	0.14	0.03	2.13	2.83	1.49	21000
061	365A - 363D	1.6250	3.5433	1.6563	1.9689	0.14	0.03	2.17	3.31	1.83	23900
062	22168 - 22325D	1.6875	3.2500	1.3750	1.7500	0.09	0.03	2.05	2.99	1.36	18200
063	25578 - 25520D	1.6875	3.2650	1.8750	2.2500	0.09	0.03	2.09	3.03	1.74	24900
064	18685 - 18620D	1.7500	3.1250	1.3125	1.6249	0.11	0.03	2.13	2.91	1.56	14500
065	25580 - 25520D	1.7500	3.2650	1.8750	2.2500	0.14	0.03	2.24	3.03	1.74	24900
066	25581 - 25520D	1.7500	3.2650	1.8750	2.2500	0.02	0.03	2.01	3.03	1.74	24900

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
067	25582 - 25520D	1.7500	3.2650	1.8750	2.2500	0.02	0.03	2.36	3.03	1.74	24900
068	355 - 353D	1.7500	3.5480	1.7500	2.0000	0,09	0.03	2.13	3.23	1.91	22800
069	355A - 353D	1.7500	3.5480	1.7500	2.0000	0.03	0.03	2.01	3.23	1.91	22800
070	3782 - 3729D	1.7500	3.6718	2.0625	2.5625	0.14	0.03	2.28	3.46	1.73	33700
071	438 - 432D	1.7500	3.7500	2.0000	2.4376	0.14	0.03	2.24	3.43	2.05	35200
072	33885 - 33821D	1.7500	3.7500	2.0624	2.5000	0.03	0.03	2.09	3.54	1.77	35800
073	460 - 452D	1.7500	4.2500	2.1250	2.5626	0.14	0.03	2.36	3.94	1.74	37400
074	535 - 533D	1.7500	4.3750	2.5000	3.1250	0.14	0.06	2.36	3.94	1.97	47400
075	18690 - 18620D	1.8125	3.1250	1.3125	1.6249	0.11	0.03	2.20	2.91	1.56	14500
076	25592 - 25520D	1.8125	3.2650	1.8750	2.2500	0.14	0.03	2.28	3.03	1.74	24900
077	359A - 353D	1.8125	3.5480	1.7500	2.0000	0.14	0.03	2.24	3.23	1.91	22800
078	359S - 353D	1.8125	3.5480	1.7500	2.0000	0.09	0.03	2.17	3.23	1.91	22800
079	3777 - 3729D	1.8125	3.6718	2.0625	2.5625	0.14	0.03	2.36	3.46	1.73	33700
080	18610 - 18620D	1.8125	3.7500	2.0000	2.4376	0.11	0.03	2.18	2.93	1.56	14500
081	436 - 432D	1.8125	3.7500	2.0000	2.4376	0.14	0.03	2.32	3.43	2.05	35200
082	369A - 363D	1.8750	3.5433	1.6563	1.9689	0.14	0.03	2.36	3.31	1.83	23900
083	369S - 363D	1.8750	3.5433	1.6563	1.9689	0.09	0.03	2.24	3.31	1.83	23900
084	3778 - 3729D	1.8750	3.6718	2.0625	2.5625	0.25	0.03	2.64	3.46	1.73	33700
085	3779 - 3729D	1.8750	3.6718	2.0625	2.5625	0.14	0.03	2.40	3.46	1.73	33700
086	386A - 384ED	1.8750	3.9370	1.5620	1.9370	0.03	0.03	2.20	3.66	1.65	25400
087	386A - 384D	1.8750	3.9370	1.6875	2.0625	0.03	0.03	2.20	3.66	1.65	25400
088	463 - 452D	1.8750	4.2500	2.1250	2.5626	0.19	0.03	2.56	3.94	1.74	37400

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
089	467 - 452D	1.8750	4.2500	2.1250	2.5626	0.03	0.03	2.24	3.94	1.74	23900
090	365S - 363D	1.9375	3.5433	1.6563	1.9689	0.03	0.03	2.17	3.31	1.83	23900
091	3781 - 3729D	1.9375	3.6718	2.0625	2.5625	0.14	0.03	2.44	3.46	1.73	33700
092	L305649 - L305610D	2.0000	3.1875	1.3750	1.6876	0.06	0.03	2.28	3.03	1.64	16900
093	368 - 363D	2.0000	3.5433	1.6563	1.9689	0.06	0.03	2.28	3.31	1.83	23900
094	368A - 363D	2.0000	3.5433	1.6563	1.9689	0.14	0.03	2.44	3.31	1.83	23900
095	370A - 363D	2.0000	3.5433	1.6563	1.9689	0.20	0.03	2.56	3.31	1.83	23900
096	3775 - 3729D	2.0000	3.6718	2.0625	2.5625	0.03	0.03	2.28	3.46	1.73	33700
097	3780 - 3729D	2.0000	3.6718	2.0625	2.5625	0.14	0.03	2.52	3.46	1.73	33700
098	3784 - 3729D	2.0000	3.6718	2.0625	2.5625	0.25	0.03	2.76	3.46	1.73	33700
099	33889 - 33821D	2.0000	3.7500	2.0624	2.5000	0.14	0.03	2.52	3.54	1.77	35800
100	385A - 384ED	2.0000	3.9370	1.5620	1.9370	0.09	0.03	2.40	3.66	1.65	25400
101	385AX - 384ED	2.0000	3.9370	1.5620	1.9370	0.03	0.03	2.28	3.66	1.65	25400
102	375 - 372XD	2.0000	3.9370	1.5626	2.0000	0.09	0.03	2.36	3.54	1.73	24400
103	385A - 384D	2.0000	3.9370	1.6875	2.0625	0.09	0.03	2.40	3.66	1.65	25400
104	385AX - 384D	2.0000	3.9370	1.6875	2.0625	0.03	0.03	2.28	3.66	1.65	25400
105	455 - 452D	2.0000	4.2500	2.1250	2.5626	0.03	0.03	2.36	3.94	1.74	37400
106	455S - 452D	2.0000	4.2500	2.1250	2.5626	0.14	0.03	2.56	3.94	1.74	37400
107	398 - 394DC	2.0000	4.3307	1.8125	2.0625	0.03	0.03	2.44	4.11	1.45	27700
108	537 - 533D	2.0000	4.3750	2.5000	3.1250	0.14	0.06	2.56	3.94	1.97	47400
109	555 - 552D	2.0000	4.8750	2.5000	3.1250	0.09	0.06	2.60	4.53	1.69	52600
110	3685 - 363D	2.0312	3.5433	1.6563	1.9689	0.08	0.03	2.32	3.31	1.83	23900

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
111	3767 - 3729	2.0625	3.6718	2.0625	3.6718	0.09	0.03	2.48	3.46	1.73	33700
112	33890 - 33821D	2.0625	3.7500	2.0624	2.5000	0.06	0.03	2.40	3.54	1.77	35800
113	33891 - 33821D	2.0625	3.7500	2.0624	2.5000	0.14	0.03	2.60	3.54	1.77	35800
114	468 - 452D	2.0625	4.2500	2.1250	2.5626	0.06	0.03	2.44	3.94	1.74	37400
115	33893 - 33821D	2.1250	3.7500	2.0624	2.5000	0.06	0.03	2.48	3.54	1.77	35800
116	389A - 384ED	2.1250	3.9370	1.5620	1.9370	0.03	0.03	2.40	3.66	1.65	25400
117	389A - 384D	2.1250	3.9370	1.6875	2.0625	0.03	0.03	2.40	3.66	1.65	25400
118	456 - 452D	2.1250	4.2500	2.1250	2.5626	0.14	0.03	2.68	3.94	1.74	37400
119	539 - 533D	2.1250	4.3750	2.5000	3.1250	0.14	0.06	2.68	3.94	1.97	47400
120	539A - 533D	2.1250	4.3750	2.5000	3.1250	0.22	0.06	2.83	3.94	1.97	47400
121	557-S - 552D	2.1250	4.8750	2.5000	3.1250	0.14	0.06	2.80	4.53	1.69	52600
122	636 - 632D	2.1250	5.3750	3.0000	3.7500	0.14	0.06	2.87	4.92	1.61	64100
123	L507949 - L507914D	2.2500	3.6250	1.3437	1.6563	0.06	0.03	2.56	3.39	1.50	16200
124	387 - 384ED	2.2500	3.9370	1.5620	1.9370	0.09	0.03	2.60	3.66	1.65	25400
125	387A - 384ED	2.2500	3.9370	1.5620	1.9370	0.14	0.03	2.72	3.66	1.65	25400
126	387-S - 384ED	2.2500	3.9370	1.5620	1.9370	0.03	0.03	2.48	3.66	1.65	25400
127	387AS - 384ED	2.2500	3.9370	1.5620	1.9370	0.20	0.03	2.83	3.66	1.65	25400
128	387 - 384D	2.2500	3.9370	1.6875	2.0625	0.09	0.03	2.60	3.66	1.65	25400
129	387A - 384D	2.2500	3.9370	1.6875	2.0625	0.14	0.03	2.72	3.66	1.65	25400
130	387-S - 384D	2.2500	3.9370	1.6875	2.0625	0.03	0.03	2.48	3.66	1.65	25400
131	387AS - 384D	2.2500	3.9370	1.6875	2.0625	0.20	0.03	2.83	3.66	1.65	25400
132	462 - 452D	2.2500	4.2500	2.1250	2.5626	0.09	0.03	2.64	3.94	1.74	37400

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Size code	Part number [1]	d	D	T	B	R [2]	r [2]	d_a	D_a	K factor	Basic dynamic load ratings (lb.)
	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
133	469 - 452D	2.2500	4.2500	2.1250	2.5626	0.14	0.03	2.76	3.94	1.74	37400
134	390 - 394D	2.2500	4.3307	1.8125	2.0625	0.09	0.03	2.76	4.11	1.45	27700
135	33225 - 33462D	2.2500	4.6250	2.1250	2.6250	0.14	0.03	2.91	4.41	1.34	38300
136	555-S - 552D	2.2500	4.8750	2.5000	3.1250	0.14	0.06	2.87	4.53	1.69	52600
137	635 - 632D	2.2500	5.3750	3.0000	3.7500	0.14	0.06	2.95	4.92	1.61	64200
138	78225 - 78549D	2.2500	5.5000	2.0395	3.0625	0.14	0.06	3.27	5.16	0.67	55200
139	28985 - 28921D	2.3750	3.9370	1.7500	2.1874	0.14	0.03	3.87	3.78	1.37	29500
140	558 - 552D	2.3750	4.8750	2.5000	3.1250	0.09	0.06	2.87	4.53	1.69	52600
141	392 - 394D	2.4375	4.3307	1.8125	2.0625	0.03	0.03	2.76	4.11	1.45	27700
142	28995 - 28921D	2.4700	3.9370	1.7500	2.1874	0.14	0.03	2.95	3.78	1.37	29500
143	L610549 - L610510D	2.5000	3.7188	1.3750	1.6874	0.06	0.03	2.80	3.58	1.38	18700
144	390A - 394D	2.5000	4.3307	1.8125	2.0625	0.06	0.03	2.87	4.11	1.45	27700
145	395 - 394D	2.5000	4.3307	1.8125	2.0625	0.14	0.03	3.03	4.11	1.45	27700
146	29585 - 29526D	2.5000	4.4375	1.6875	2.1875	0.14	0.03	3.03	4.13	1.27	24400
147	29586 - 29526D	2.5000	4.4375	1.6875	2.1875	0.06	0.03	2.87	4.13	1.27	24400
148	477 - 472D	2.5000	4.7244	2.1250	2.5626	0.03	0.03	2.87	4.49	1.52	39500
149	483 - 472D	2.5000	4.7244	2.1250	2.5626	0.14	0.03	3.07	4.49	1.52	39500
150	559 - 552D	2.5000	4.8750	2.5000	3.1250	0.14	0.06	3.07	4.53	1.69	52600
151	565 - 563D	2.5000	5.0000	2.5625	3.1875	0.14	0.06	3.15	4.69	1.61	50000
152	639 - 632D	2.5000	5.3750	3.0000	3.7500	0.14	0.06	3.19	4.92	1.61	64100
153	745S - 742D	2.5000	6.1250	3.3750	4.0000	0.14	0.06	3.31	5.63	1.80	87100
154	395A - 394D	2.6250	4.3307	1.8125	2.0625	0.03	0.03	2.87	4.11	1.45	27700

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
155	395-S - 394D	2.6250	4.3307	1.8125	2.0625	0.14	0.03	3.11	4.11	1.45	27700
156	29590 - 29526D	2.6250	4.4375	1.6875	2.1875	0.14	0.03	3.15	4.13	1.27	24400
157	560 - 552D	2.6250	4.8750	2.5000	3.1250	0.14	0.06	3.19	4.53	1.69	52600
158	641 - 632D	2.6250	5.3750	3.0000	3.7500	0.14	0.06	3.27	4.92	1.61	64100
159	399A - 394D	2.6875	4.3307	1.8125	2.0625	0.09	0.03	3.07	4.11	1.45	27700
160	399AS - 394D	2.6875	4.3307	1.8125	2.0625	0.20	0.03	3.27	4.11	1.445	27700
161	33269 - 33462D	2.6875	4.6250	2.1250	2.6250	0.14	0.03	3.23	4.41	1.34	38300
162	480 - 472D	2.6875	4.7244	2.1250	2.5626	0.14	0.03	3.23	4.49	1.52	12600
163	560-S - 552D	2.6875	4.8750	2.5000	3.1250	0.14	0.06	3.27	4.53	1.69	52600
164	570 - 563D	2.6875	5.0000	2.5625	3.1875	0.14	0.06	3.27	4.69	1.61	50000
165	29675 - 29622D	2.7500	4.4995	1.8125	2.3125	0.06	0.03	3.15	4.29	1.20	30800
166	33275 - 33462D	2.7500	4.6250	2.1250	2.6250	0.14	0.03	3.31	4.41	1.34	38300
167	482 - 472D	2.7500	4.7244	2.1250	2.5626	0.14	0.03	3.27	4.49	1.52	39600
168	47487 - 47420D	2.7500	4.7244	2.3125	2.8125	0.14	0.03	3.31	4.49	1.62	49300
169	566 - 563D	2.7500	5.0000	2.5625	3.1875	0.14	0.06	3.35	4.69	1.61	50000
170	643 - 632D	2.7500	5.3750	3.0000	3.7500	0.14	0.06	3.39	4.92	1.61	64100
171	655 - 654D	2.7500	6.0000	3.0000	3.7500	0.14	0.06	3.46	5.55	1.43	67900
172	745A - 742D	2.7500	6.1250	3.3750	4.0000	0.14	0.06	3.46	5.63	1.80	87000
173	835 - 834D	2.7500	6.7500	3.9375	4.9375	0.14	0.03	3.58	6.10	1.95	111500
174	34274 - 34478D	2.7540	4.78112	1.5000	2.0626	0.08	0.03	3.19	4.57	1.30	28500
175	33281 - 33462D	2.8125	4.6250	2.1250	2.6250	0.14	0.03	3.35	4.41	1.34	38300
176	47490 - 47420D	2.8125	4.7244	2.3125	2.8125	0.14	0.03	3.39	4.49	1.62	49300

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
177	567A - 563D	2.8125	5.0000	2.5625	3.1875	0.14	0.06	3.39	4.69	1.61	50000
178	495-S - 493D	2.8125	5.3750	2.1250	2.7500	0.14	0.03	3.46	5.12	1.31	42800
179	644 - 632D	2.8125	5.3750	3.0000	3.7500	0.14	0.06	3.43	4.92	1.61	64100
180	29685 - 29622D	2.8750	4.4995	1.8125	2.3125	0.14	0.03	3.39	4.29	1.20	30800
181	33287 - 33462D	2.8750	4.6250	2.1250	2.6250	0.14	0.03	3.43	4.41	1.34	38300
182	567 - 563D	2.8750	5.0000	2.5625	3.1875	0.14	0.06	3.46	4.69	1.61	50000
183	659 - 654D	2.8750	5.0000	2.5625	3.1875	0.14	0.06	3.46	4.69	1.43	67900
184	576 - 572D	2.8750	5.5115	2.6250	3.2500	0.14	0.03	3.54	5.24	1.45	51100
185	657 - 654D	2.8750	6.0000	3.0000	3.7500	0.14	0.06	3.58	5.55	1.43	67900
186	744 - 742D	2.8750	6.1250	3.3750	4.0000	0.14	0.06	3.58	5.63	1.80	87000
187	762 - 752D	2.8750	6.3750	3.3750	4.1250	0.14	0.06	3.62	5.91	1.71	89700
188	29688 - 29622D	2.9062	4.4995	1.8125	2.3125	0.06	0.03	3.27	4.29	1.20	30800
189	568 - 563D	2.9375	5.5115	2.6250	3.2500	0.14	0.03	3.58	5.24	1.61	50000
190	577 - 572D	3.0000	4.1563	0.9374	1.2500	0.06	0.03	3.27	4.02	1.45	51100
191	LL714649 - LL714610D	3.0000	4.3125	1.3750	1.6874	0.06	0.03	3.31	4.13	1.24	30800
192	L814749 - L814710D	3.0000	4.7812	1.5000	2.0626	0.08	0.03	3.39	4.57	1.16	19400
193	34300 - 34478D	3.0000	4.7812	1.5000	2.0626	0.14	0.03	3.50	4.57	1.30	28600
194	34301 - 34478D	3.0000	5.3750	2.1250	2.7500	0.14	0.03	3.62	5.12	1.30	28600
195	485A - 493D	3.0000	5.3750	2.1250	2.7500	0.25	0.03	3.86	5.12	1.31	42800
196	495AX - 493D	3.0000	5.3750	2.1250	2.7500	0.25	0.03	3.86	5.12	1.31	42800
197	575 - 572D	3.0000	5.5115	2.6250	3.2500	0.14	0.03	3.62	5.24	1.45	29800

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
198	575-S - 572D	3.0000	5.5115	2.6250	3.2500	0.27	0.03	3.90	5.24	1.45	29800
199	590A - 592D	3.0000	6.0000	2.5000	3.2500	0.14	0.03	3.74	5.67	1.32	59600
200	659 - 654D	3.0000	6.0000	3.0000	3.7500	0.14	0.06	3.66	5.55	1.43	67900
201	748-S - 742D	3.0000	6.1250	3.3750	4.0000	0.14	0.06	3.66	5.63	1.80	87000
202	755 - 752D	3.0000	6.3750	3.3750	4.1250	0.14	0.06	3.74	5.91	1.71	89700
203	837 - 934D	3.0000	6.7500	3.9375	4.9375	0.03	0.03	3.54	6.10	1.95	111500
204	843 - 834D	3.0000	6.7500	3.9375	4.9375	0.25	0.03	3.98	6.10	1.95	111500
205	HH221430 - HH221410D	3.0000	7.5000	4.1250	5.0000	0.14	0.06	3.98	7.05	1.74	14600
206	34306 - 34478D	3.0625	4.7812	1.5000	2.0626	0.14	0.03	3.54	4.57	1.30	28500
207	34307 - 34478D	3.0625	4.7812	1.5000	2.0626	0.25	0.03	3.78	4.57	1.30	28500
208	495AS - 493D	3.0625	5.3750	2.1250	2.7500	0.14	0.03	3.66	5.12	1.31	42800
209	661 - 654D	3.1250	6.0000	3.0000	3.7500	0.14	0.06	3.78	5.55	1.43	67900
210	756A - 752D	3.1250	6.3750	3.3750	4.1250	0.31	0.06	4.17	5.91	1.71	89700
211	HH221431 - HH221410D	3.1250	7.5000	4.1250	5.0000	0.14	0.06	4.06	7.05	1.74	146000
212	496 - 493D	3.1875	5.3750	2.1250	2.7500	0.14	0.03	3.74	5.12	1.31	42800
213	581 - 572D	3.1875	5.5115	2.6250	3.2500	0.14	0.03	3.78	5.24	1.45	51100
214	740 - 742D	3.1875	6.1250	3.3750	4.0000	0.20	0.06	3.98	5.63	1.80	87000
215	L116149 - L116110D	3.2500	4.5625	1.5626	1.8750	0.06	0.03	3.54	4.37	1.90	15000
216	495 - 493D	3.2500	5.3750	2.1250	2.7500	0.14	0.03	3.82	5.12	1.31	42800
217	580 - 572D	3.2500	5.5115	2.6250	3.2500	0.14	0.03	3.86	5.24	1.45	51100

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
218	582 - 572D	3.2500	5.5115	3.6250	3.2500	0.27	0.03	4.09	5.24	1.45	51100
219	595 - 592D	3.2500	6.0000	2.5000	3.2500	0.14	0.03	3.94	5.67	1.32	59600
220	663 - 654D	3.2500	6.0000	3.0000	3.7500	0.14	0.06	3.90	5.55	1.43	67900
221	749A - 742D	3.2500	6.1250	3.3750	4.0000	0.14	0.06	3.90	5.63	1.80	87000
222	750A - 742D	3.2500	6.1250	3.3750	4.0000	0.27	0.06	4.17	5.63	1.80	87000
223	757 - 752D	3.2500	6.3750	3.3750	4.1250	0.14	0.06	3.94	5.91	1.71	89700
224	842 - 834D	3.2500	6.7500	3.9375	4.9375	0.14	0.03	3.98	6.10	1.95	111500
225	498 - 493D	3.3125	5.3750	2.1250	2.7500	0.14	0.03	3.86	5.12	1.31	42800
226	497 - 493D	3.3750	5.3750	2.1250	2.7500	0.14	0.03	3.90	5.12	1.31	42800
227	497A - 493D	3.3750	5.3750	2.1250	2.7500	0.25	0.03	4.13	5.12	1.31	42800
228	596 - 592D	3.3750	6.0000	2.5000	3.2500	0.14	0.03	4.02	5.67	1.32	59600
229	665 - 654D	3.3750	6.0000	3.0000	3.7500	0.14	0.06	4.02	5.55	1.43	67900
230	665A - 654D	3.3750	6.0000	3.0000	3.7500	0.25	0.06	4.21	5.55	1.43	67900
231	758 - 752D	3.3750	6.3750	3.3750	4.1250	0.14	0.08	4.06	5.91	1.71	89700
232	677 - 672D	3.3750	6.6250	2.7500	3.6250	0.14	0.03	4.13	6.30	1.24	73000
233	841 - 834D	3.3750	6.7500	3.9375	4.9375	0.14	0.03	4.09	6.10	1.95	111500
234	L217849 - L217810D	3.5000	4.8750	1.6875	1.9999	0.06	0.03	3.82	4.69	1.77	25900
235	42350 - 42587D	3.5000	5.8750	2.0625	2.6249	0.12	0.03	4.09	5.63	1.19	45300
236	593 - 592D	3.5000	6.0000	2.5000	3.2500	0.14	0.03	4.09	5.67	1.32	59600
237	593A - 592D	3.5000	6.0000	2.5000	3.2500	0.25	0.03	4.33	5.67	1.32	59600
238	759 - 752D	3.500	6.3750	3.3750	4.1250	0.14	0.06	4.17	5.91	1.71	89700
239	766 - 752D	3.5000	6.3750	3.3750	4.1250	0.28	0.06	4.45	5.91	1.71	89700

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
240	679 - 672D	3.5000	6.6250	2.7500	3.6250	0.14	0.03	4.21	6.30	1.24	68200
241	850 - 834D	3.5000	6.7500	3.9375	4.9375	0.14	0.03	4.17	6.10	1.95	111500
242	855 - 854D	3.5000	7.5000	4.0000	5.0000	0.31	0.06	4.65	6.85	1.74	125400
243	HH221434 - HH221410D	3.5000	7.5000	4.1250	5.0000	0.31	0.06	4.72	7.05	1.74	146000
244	760 - 752D	3.5625	6.3750	3.3750	4.1250	0.14	0.06	4.21	5.91	1.71	89700
245	598 - 592D	3.6250	6.0000	2.5000	3.2500	0.14	0.03	4.21	5.67	1.32	59600
246	598A - 592D	3.6250	6.0000	2.5000	3.2500	0.25	0.03	4.45	5.67	1.32	59600
247	681 - 672D	3.6250	6.6250	2.7500	3.6250	0.14	0.03	4.33	6.30	1.24	68200
248	681A - 672D	3.6250	6.6250	2.7500	3.6250	0.25	0.03	4.57	6.30	1.24	68200
249	778 - 774D	3.6250	7.1250	3.3750	4.1250	0.14	0.06	4.37	6.61	1.51	88200
250	857 - 854D	3.6250	7.5000	4.0000	5.0000	0.31	0.06	4.76	6.85	1.74	125400
251	42368 - 42587D	3.6875	5.8750	2.0625	2.6249	0.12	0.03	4.21	5.63	1.19	59600
252	L319249 - L319210D	3.7500	5.1250	1.5625	1.8749	0.06	0.03	4.06	4.92	1.67	26700
253	42375 - 42587D	3.7500	5.8750	2.0625	2.6249	0.12	0.03	4.25	5.63	1.19	59600
254	594 - 592D	3.7500	6.0000	2.5000	3.2500	0.14	0.03	4.33	5.67	1.32	59600
255	594A - 592D	3.7500	6.0000	2.5000	3.2500	0.20	0.03	4.45	5.67	1.32	59600
256	52375 - 52637D	3.7500	6.3750	2.4375	3.2499	0.14	0.03	4.41	6.06	1.23	61900
257	683 - 672D	3.7500	6.6250	2.7500	3.6250	0.14	0.03	4.45	6.30	1.24	68200
258	683XA - 672D	3.7500	6.6250	2.7500	3.6250	0.20	0.03	4.57	6.30	1.24	68200
259	864 - 854D	3.7500	7.5000	4.0000	5.0000	0.31	0.06	4.84	6.85	1.74	125400

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Size code	Part number [1]	d	D	T	B	R [2]	r [2]	d_a	D_a	K factor	Basic dynamic load ratings (lb.)
	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
260	HH221440 - HH221410D	3.7500	7.5000	4.1250	5.0000	0.31	0.06	4.92	7.05	1.74	14600
261	42381 - 42587D	3.8125	5.8750	2.0625	2.6249	0.14	0.03	4.33	5.63	1.19	45300
262	52387 - 52637D	3.8750	6.3750	2.4375	3.2499	0.14	0.03	4.49	6.06	1.23	61900
263	685 - 672D	3.8750	6.6250	2.7500	3.6250	0.14	0.03	4.57	6.30	1.24	68200
264	HH221442 - HH221410D	3.8750	7.5000	4.1250	5.0000	0.14	0.06	4.69	7.05	1.74	146000
265	52393 - 52637D	3.9375	2.4375	3.2499	6.3750	0.14	0.03	4.57	6.06	1.23	61900
266	L521945 - L521910D	4.0000	1.5625	1.9375	5.7500	0.06	0.03	4.41	5.55	1.49	26900
267	52400 - 52637D	4.0000	6.3750	2.4375	3.2499	0.14	0.03	4.61	6.06	1.23	61900
268	687 - 672D	4.0000	6.6250	2.7500	3.6250	0.14	0.03	4.65	6.30	1.24	68200
269	780 - 774D	4.0000	7.1250	3.3750	4.1250	0.14	0.06	4.69	6.61	1.51	88200
270	861 - 854D	4.0000	7.5000	4.0000	5.0000	0.31	0.06	5.08	6.85	1.74	125400
271	HH221449 - HH221410D	4.0000	7.5000	4.1250	5.0000	0.31	0.06	5.16	7.05	1.74	146000
272	HH221449A - H221410D	4.0000	7.5000	4.1250	5.0000	0.14	0.06	4.80	7.05	1.74	146000
273	941 - 932D	4.0000	8.3750	3.3750	4.0000	0.14	0.03	5.87	7.52	1.79	142500
274	782 - 773D	4.1250	7.0866	3.3750	4.1250	0.14	0.03	4.80	6.61	1.51	88200
275	782 - 774D	4.1250	7.1250	3.3750	4.1250	0.14	0.06	4.80	6.61	1.51	88200
276	786 - 774D	4.1250	7.1250	3.3750	4.1250	0.25	0.06	5.04	6.61	1.51	88200
277	787 - 774D	4.1250	7.1250	3.3750	4.1250	0.28	0.06	5.08	6.61	1.51	88200
278	71412 - 71751D	4.1250	7.5000	3.1875	4.1875	0.14	0.06	4.88	7.13	1.40	100000

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
279	56418 - 56650D	4.1875	6.5000	2.5000	3.2500	0.14	0.03	4.80	6.26	1.18	62700
280	L521949 - L521910D	4.2500	5.7500	1.5625	1.9375	0.06	0.03	4.57	5.55	1.49	26900
281	LM522546 - LM522510D	4.2500	6.2987	2.3125	2.9375	0.14	0.03	4.80	6.06	1.45	54200
282	56425 - 56650D	4.2500	6.5000	2.5000	3.2500	0.14	0.03	4.84	6.26	1.18	62700
283	71425 - 71751D	4.2500	7.5000	3.1875	4.1875	0.14	0.06	4.96	7.13	1.40	100000
284	L623149 - L623110D	4.5000	6.0000	1.5000	1.8750	0.06	0.03	4.84	5.79	1.41	28000
285	64450 - 64700D	4.5000	7.0000	2.7500	3.6250	0.14	0.03	5.16	6.77	1.13	75900
286	71450 - 71751D	4.5000	7.5000	3.1875	4.1875	0.14	0.06	5.20	7.13	1.40	100000
287	938 - 932D	4.5000	8.3750	4.6250	5.6250	0.28	0.06	5.55	7.60	1.79	142900
288	HH224346 - HH224310D	4.5000	8.3750	4.6250	5.6250	0.28	0.06	5.63	7.94	1.79	185600
289	71453 - 71751D	4.5310	7.5000	3.1875	4.1875	0.14	0.06	5.24	7.13	1.40	100000
290	L624549 - L624514D	4.7500	6.3750	2.1250	2.5000	0.06	0.03	5.08	6.14	1.34	29600
291	L225842 - L225812D	4.7500	6.6919	1.9375	2.3125	0.06	0.04	5.16	6.46	1.76	39900
292	M224749 - M224710D	4.7500	6.8750	2.4375	3.0625	0.14	0.03	5.31	6.61	1.76	68000
293	48286 - 48220D	4.8750	7.1875	2.8750	3.3750	0.14	0.03	5.47	6.93	1.91	74000
294	L225849 - L225812D	5.0000	6.6919	1.9375	2.3125	0.06	0.04	5.35	6.46	1.76	39900
295	48290 - 48220D	5.0000	7.1875	2.8750	3.3750	0.14	0.03	5.55	6.93	1.91	74000

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
296	67388 - 67322D	5.0000	7.7500	3.3750	4.0000	0.14	0.03	5.67	7.48	1.24	54900
297	95500 - 95927D	5.0000	9.2500	4.5000	5.6250	0.25	0.06	6.06	8.54	1.58	153200
298	799 - 792D	5.0625	8.1250	3.2500	4.2500	0.13	0.03	5.75	7.80	1.27	97800
299	67389 - 67322D	5.1250	7.7500	3.3750	4.0000	0.14	0.03	5.75	7.48	1.24	54900
300	L327249 - L327210D	5.2500	6.9688	1.8750	2.2500	0.06	0.03	5.59	6.73	1.68	41200
301	48385 - 48320D	5.2500	7.5000	2.8750	3.3750	0.14	0.03	5.83	7.24	1.82	75000
302	67390 - 67322D	5.2500	7.7500	3.3750	4.0000	0.14	0.03	5.87	7.48	1.24	54900
303	67391 - 67322D	5.2500	7.7500	3.3750	4.0000	0.31	0.03	6.18	7.48	1.24	54900
304	67390 - 67323D	5.2500	7.7500	3.6250	4.2500	0.14	0.03	5.87	7.48	1.24	54900
305	67390 - 67325D	5.2500	7.8750	3.3750	4.0000	0.14	0.03	5.87	7.52	1.24	54900
306	95525 - 95927D	5.2500	9.2500	4.5000	5.6250	0.38	0.06	6.54	8.54	1.58	153000
307	48393 - 48320D	5.3750	7.5000	2.8750	3.3750	0.14	0.03	5.94	7.24	1.82	75000
308	74550 - 74851D	5.5000	8.5000	3.1875	4.1875	0.14	0.06	6.22	8.19	1.20	98800
309	73551 - 73876D	5.5000	8.7500	2.1250	2.9800	0.14	0.09	6.14	8.15	1.34	42400
310	898 - 892D	5.5000	9.0000	3.8750	4.8750	0.14	0.06	6.30	8.50	1.39	142500
311	99550 - 99102D	5.5000	10.0000	4.3750	5.8750	0.28	0.06	6.69	9.37	1.43	156800
312	HH234031 - HH234011D	5.5000	12.1250	6.1250	7.8750	0.38	0.09	7.09	11.24	1.79	333700
313	48684 - 48620D	5.6250	7.8750	2.8750	3.4376	0.31	0.03	6.54	7.60	1.74	74600
314	48685 - 48620D	5.6250	7.8750	2.8750	3.4376	0.14	0.03	6.22	7.60	1.74	74600
315	36690 - 36620D	5.7500	7.6250	2.1250	2.5624	0.06	0.03	6.10	7.40	1.59	58600
316	36691 - 36620D	5.7500	7.6250	2.1250	2.5624	0.19	0.03	6.38	7.40	1.59	58600

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
317	82576 - 82951D	5.7500	9.5000	4.1875	5.1875	0.14	0.06	6.54	8.90	1.32	141300
318	99575 - 99101D	5.7500	10.0000	4.5000	6.0000	0.28	0.06	6.89	9.37	1.43	156800
319	HM231149 - HM231111D	5.8750	9.3125	4.1875	5.1875	0.14	0.06	6.57	8.82	1.83	148200
320	HM231149 - HM231116D	5.8750	9.5000	4.1875	5.1875	0.14	0.06	6.57	8.82	1.83	148200
321	99587 - 99102D	5.8750	10.0000	4.3750	5.8750	0.28	0.06	7.01	9.37	1.43	156800
322	LM330448 - LM330410D	6.0000	8.0000	3.1250	3.6250	0.13	0.03	6.54	7.76	1.68	78300
323	M231649 - M231610D	6.0000	8.7500	3.0000	3.9374	0.14	0.03	6.65	8.39	1.76	94300
324	99600 - 99101D	6.0000	10.0000	4.5000	6.0000	0.28	0.06	7.13	9.37	1.43	156800
325	EE107060 - 107105D	6.0000	10.5625	4.9375	6.3125	0.25	0.06	7.13	9.82	1.51	205200
326	EE450601 - 451215D	6.0000	12.1250	5.7500	7.8750	0.38	0.09	7.44	10.82	1.79	288200
327	HH234048 - HH234011D	6.0000	12.1250	6.1250	7.8750	0.38	0.09	7.52	11.24	1.79	333700
328	L433749 - L433710D	6.5000	8.5000	1.8750	2.3126	0.06	0.03	6.85	8.23	1.60	50300
329	46790 - 46720D	6.5000	8.8750	2.7500	3.3750	0.14	0.03	7.13	8.58	1.52	78200
330	86650 - 86100XD	6.5000	10.0000	3.0000	4.0000	0.19	0.06	7.28	9.41	1.58	106700
331	M235145 - M235113D	6.5000	10.0000	3.0000	4.0000	0.19	0.06	7.28	9.45	1.83	119700
332	94687 - 94114D	6.8750	11.3750	4.3750	5.6250	0.28	0.06	8.03	10.71	1.25	173800

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
333	67790 - 67720D	7.0000	9.7500	3.3125	4.0625	0.14	0.03	7.64	9.45	1.33	66800
334	HM237545 - HM237510D	7.0000	11.3750	4.3750	5.6250	0.28	0.06	8.07	10.68	1.83	211800
335	94700 - 94118D	7.0000	11.7500	4.3750	5.6250	0.28	0.06	8.15	10.71	1.25	173800
336	EE420701 - 421451D	7.0000	14.5000	5.3750	7.6250	0.05	0.06	9.09	13.16	1.445	320200
337	LL537649 - LL537610D	7.2500	9.3125	1.6250	2.1874	0.06	0.03	7.64	9.06	1.45	49200
338	M238849 - M238810D	7.3750	10.6250	3.6875	4.6875	0.14	0.06	8.07	10.08	1.76	151000
339	87737 - 87112D	7.3750	11.1250	3.1250	4.2500	0.14	0.06	8.15	10.50	1.45	119700
340	H239649 - H239612D	7.3750	12.6250	5.4375	7.3125	0.22	0.06	8.43	11.73	1.83	303000
341	67885 - 67820D	7.5000	10.5000	3.3125	4.0625	0.14	0.03	8.23	10.20	1.22	115900
342	87750 - 87112D	7.5000	11.1250	3.1250	4.2500	0.14	0.06	8.23	10.50	1.45	119700
343	EE420751 - 421451D	7.5000	14.5000	5.3750	7.6250	0.25	0.06	8.94	13.16	1.45	320200
344	87762 - 87112D	7.6250	11.1250	3.1250	4.2500	0.14	0.06	8.31	10.50	1.41	119700
345	L540049 - L540010D	7.7500	10.0000	1.8750	2.4374	0.06	0.03	8.15	9.72	1.47	59600
346	LM739749 - LM739710D	7.7500	10.1250	2.6250	3.3750	0.14	0.03	8.39	9.88	1.31	88800
347	LM241149 - LM241110D	8.0000	10.8750	2.8750	3.5624	0.14	0.03	8.62	10.51	1.83	112100
348	M241547 - M241510D	8.0000	11.5000	4.0000	4.9376	0.14	0.06	8.70	10.98	1.76	122600

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	Cone-cup	Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter: Shaft	Recommended shoulder diameter: Housing		Two row radial
349	EE420801 - 421451D	8.0000	14.5000	5.3750	7.6250	0.13	0.06	9.06	13.16	1.45	320200

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3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing dimensions (and dynamic load ratings, see 3.4) shall conform to the requirements specified in table I for each of the bearing part numbers. The listed dimensions conform to the requirements specified in American Bearing Manufacturers Association (ABMA) Standard 19.2, "Tapered Roller Bearings - Radial Inch Design". The bearing size shall be specified in the acquisition order (see 7.3(b)). For any unlisted bearing size codes, the associated dimensional and dynamic load rating requirements should also be specified in the acquisition order.

3.2 Materials.

3.2.1 Cones (inner rings), cups (outer rings), and rollers. The bearing cones, cups, and rollers shall be made of case carburized or through-hardened steel produced in accordance with the ASTM International (ASTM) A295/A295M, "Standard Specification for High-Carbon Anti-Friction Bearing Steel", or ASTM A534, "Standard Specification for Carburizing Steels for Anti-Friction Bearings". The steel shall show a fine fracture grain size in accordance with ASTM E112, "Standard Test Methods for Determining Average Grain Size". Material hardness shall be no less than Rockwell hardness number of 58 on Rockwell C scale (HRC) and no more than 64 HRC as defined in ASTM E18, "Standard Test Methods for Rockwell Hardness of Metallic Materials".

3.2.2 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The cages shall be made from carbon steel (one piece stamped). Materials shall operate from -65 to 230 °F (-53.9 to 110 °C).

3.3 Tolerance class. The tolerance limits for bearings shall conform to tolerance class 4 as tabulated in ABMA Standard 19.2. Allowable tolerances for bearing components and assembled bearings are listed in tables II through V.

TABLE II. Cone bore tolerance.

Cone bore (d)			
Size range		Tolerance	
Over	Inclusive	Plus	Minus
0.0000	3.0000	5	0
3.0000	6.0000	10	0

Note: Allowable tolerances are in 0.0001 inch.

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TABLE III. Cup diameter tolerance.

Cup diameter (D)			
Size range		Tolerance	
Over	Inclusive	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE IV. Bearing width tolerance.

Bearing width (T)			
Bore size range		Tolerance	
Over	Inclusive	Plus	Minus
0.0000	4.0000	160	0
4.0000	12.0000	280	200

Note: Allowable tolerances are in 0.0001 inch.

TABLE V. Assembled bearing tolerance.

Assembled bearing maximum radial runout		
Cup outside diameter (D)		Tolerance
Over	Inclusive	
0.0000	24.0000	20

Note: Allowable tolerances are in 0.0001 inch.

3.4 Dynamic load rating. The bearing dynamic load rating shall conform to the requirements specified in table I for each bearing size code. The listed ratings conform to the requirements specified in ABMA Standard 11, "Load Ratings and Fatigue Life for Roller Bearings".

3.5 Lubrication. The bearings shall be furnished without lubrication.

3.6 Contact angle. All bearings are normal angle bearings having a contact angle between 10 and 19 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Foreign acquisition restrictions. Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 252.225, Clause 252.225.7016, of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

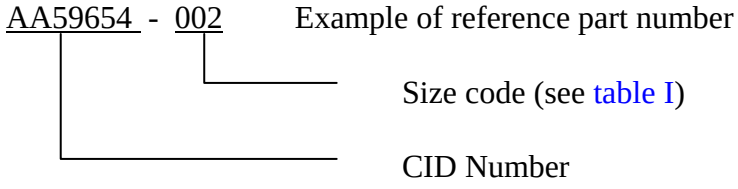
5.2 Market acceptability. The products offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified in the acquisition order, the bearings shall be preserved, packaged, and marked in accordance with MIL-DTL-197, "Packaging of Bearings, Associated Parts and Subassemblies" (see 7.3(c)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



AA59654 - 002 indicates: Bearing bore 0.5000 inches; outside diameter 1.3775 inches; width 0.8125 inches.

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7.2 Sources of documents.

7.2.1 DFARS and FAR. Copies of DFARS and FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfarspgi/current/index.html. Electronic copies of FAR may be obtained from https://www.acquisition.gov/far/.

7.2.2 Military specifications. Copies of military specifications may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies may be obtained from https://assist.daps.dla.mil/.

7.2.3 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 2025 M Street NW, Suite 800, Washington, DC 20036. Electronic copies may be obtained from http://www.abma-dc.org/.

7.2.4 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from http://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1).
c. Preservation, packaging, and marking requirements (see 6.1).

7.4 Cross-reference information. Table VI relates the original specification slant sheets to the replacement CIDs.

TABLE VI. Federal specification to CID cross-reference.

FF-B-187B specification sheets	Replacement CID numbers	ABMA types
1	A-A-59649	TS
2	A-A-59650	TSF
3	A-A-59651	TSS
4	A-A-59652	TSSF
5	A-A-59653	TDI and TDIS
6	A-A-59654	TDO
7	A-A-59655	TDOS
8	A-A-59656	TNA (normal angle)
9	A-A-59657	TNAS (steep angle)
10	A-A-59658	TNASW
11	A-A-59659	TNASWE

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7.5 Subject term (key word) listing.

Bore
Cone
Cup
Load
Width

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - MC
Air Force - 99
DLA - GS

Review Activities:
Navy - OS
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA - GS4

(Project 3110-2011-023)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.daps.dla.mil/.

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